

• • LIVE + RECORDED • STANDARD COURSE

Python for Data Science — 8 Weeks

NumPy, Pandas, Matplotlib and the full data workflow. A structured 8-week program covering the complete Python data science stack — from raw data to actionable insights.

PRICE	DURATION	LEVEL	FORMAT	SCHEDULE
₹5,999	8 Weeks	Intermediate	Live + Recorded	Sat & Sun · 10 AM–12 PM



Prasant Mishra

Founder & Instructor, Qurious Academy

ABOUT THIS COURSE

What is Python for Data Science — 8 Weeks?

Data science in Python has a well-defined core toolkit: NumPy for numerical computation, Pandas for data manipulation, and Matplotlib/Seaborn for visualisation. These three libraries, used well, let you take a raw dataset and produce real insights. This 8-week course teaches all three — thoroughly, with real datasets, in a way that builds lasting skills rather than surface familiarity.

The course is project-driven: every two weeks builds to a real dataset project. You'll clean a messy real-world dataset, run a full exploratory data analysis on Indian census data, and complete a capstone end-to-end analysis project of your choosing. By Week 8 you'll have a Jupyter notebook portfolio that demonstrates your ability to handle data from start to finish.

The weekend format (Saturday and Sunday mornings for 8 weeks) is designed to be sustainable alongside a full-time job or university. Sessions are two hours each, and recordings are available for anything you miss or want to review. Weekly assignments with real datasets ensure you're building fluency, not just following along.

IS THIS RIGHT FOR YOU?

Who Should Enroll

✓ THIS IS FOR YOU IF...

- ✓ You know basic Python and want to transition into data science or analytics roles
- ✓ You work with data in Excel or SQL and want to move to Python for more powerful analysis
- ✓ You're a student targeting data science, ML, or analyst roles and need a strong Python data science foundation
- ✓ You want to produce publication-quality data visualisations and Jupyter notebook analyses to show employers

✗ THIS IS NOT FOR YOU IF...

- ✗ You don't know Python at all — take the Python Weekend Intensive or Masterclass first, then return to this course
- ✗ You want to train ML models — this course covers the data preparation and exploration side; for ML modelling see the AI & ML Engineering 12-Week program
- ✗ You're looking for SQL, database engineering, or big data (Spark, Databricks) — this course is Python-only and focuses on in-memory analysis with Pandas

Prerequisites: Basic Python — comfortable with variables, loops, functions, and importing libraries. No statistics background required; we introduce the relevant concepts as needed.

Eight Weeks, Four Phases, Four Projects

WEEKS 1-2**Python & NumPy Foundations**

- ▶ **Python refresher (fast-paced)** — a quick pass through the Python you'll use constantly in data science: list comprehensions, lambda functions, map/filter, and working with files
- ▶ **NumPy arrays & broadcasting** — the ndarray as Python's most important data structure for numerical computing; array creation, indexing, slicing, and the broadcasting rules that let you write clean vectorised code
- ▶ **Vectorised operations** — replacing Python loops with NumPy operations for 10–100x performance; universal functions (ufuncs) and when to reach for them
- ▶ **Random number generation & distributions** — generating random data, sampling from distributions (uniform, normal, Binomial), and the applications in simulation and statistical analysis

WEEKS 3-4**Pandas — Data Wrangling**

- ▶ **DataFrames & Series** — Pandas' two core data structures; creating them, understanding their index, and the relationship between them
- ▶ **Indexing, filtering & groupby** — .loc vs. .iloc, boolean masking, and the groupby-apply-combine paradigm for aggregating data across categories
- ▶ **Merging, joining & reshaping** — combining datasets with merge and join; pivoting and melting for reshaping data between wide and long formats
- ▶ **Handling missing data + Project** — strategies for detecting, imputing, and removing missing values; Project: receive a genuinely messy real-world dataset and clean it into analysis-ready form

WEEKS 5-6**Visualisation & EDA**

- ▶ **Matplotlib & Seaborn** — the two-layer approach to Python visualisation; Matplotlib for full control, Seaborn for statistical plots with less code; figure and axes anatomy
- ▶ **Exploratory Data Analysis workflow** — the systematic process of getting to know a new dataset: distributions, outliers, correlations, and unexpected patterns
- ▶ **Distribution plots, heatmaps, pair plots** — the visualisation types most useful for EDA; choosing the right chart for the question you're asking
- ▶ **Project: full EDA on Indian census data** — a complete exploratory analysis on a real, meaningful dataset; the full EDA workflow from loading to conclusions

WEEKS 7-8**Applied Analysis & Capstone**

- ▶ **Statistical testing basics** — hypothesis testing, p-values, t-tests, and chi-squared tests in Python; when to reach for a statistical test and how to interpret the result without being misled
- ▶ **Correlation vs. causation** — the most important distinction in data analysis; how to present correlations honestly and what additional evidence is needed for causal claims

- ▶ **Reporting with Jupyter notebooks** — structuring a notebook as a document someone else can read; markdown, clean visualisations, and the narrative arc of a data analysis report
- ▶ **Capstone: end-to-end analysis project** — you choose a dataset on a topic you care about; full EDA, visualisations, statistical analysis, and a polished Jupyter notebook report with instructor feedback

LEARNING OUTCOMES

What You'll Walk Away With

- 01 Clean, transform and analyse real datasets with Pandas**
— handling messy, real-world data at the scale and complexity that actually appears in jobs and projects

- 02 Create clear, publication-quality visualisations**
— charts that communicate your findings effectively, not just technically correct plots

- 03 Run an end-to-end EDA workflow**
— the systematic approach to exploring a new dataset: from first load to actionable conclusions

- 04 Present findings in polished Jupyter notebooks**
— a capstone notebook that serves as a portfolio piece demonstrating the complete data science workflow

WHAT'S INCLUDED

Everything You Get

- ✓ **Live sessions + recordings** — Saturday and Sunday mornings for 8 weeks; all recordings permanently accessible
- ✓ **All datasets & notebooks** — every dataset used in the course and every Jupyter notebook from the sessions, ready to run and experiment with
- ✓ **Weekly assignments** — real dataset assignments every week to build fluency; model solutions shared after submission
- ✓ **Capstone feedback** — personalised written feedback on your capstone notebook from the instructor
- ✓ **Learner's Certificate** — digital certificate of completion from Curious Academy

LOGISTICS

Schedule & Pricing

LIVE DAYS

Saturday & Sunday

TIME

10:00 AM – 12:00 PM IST

DURATION

PRICE

8 weeks · 16 sessions · 32 hrs live

₹5,999

❏ **Do I need to know statistics before starting?**

No — we introduce statistical concepts (mean, variance, distributions, hypothesis testing) as we need them. You don't need a formal statistics background.

❏ **What tool do we use — Jupyter, VS Code, Google Colab?**

We recommend Jupyter Lab locally for the best experience. Google Colab is a free alternative that requires no installation. We support both — setup instructions sent before Week 1.

❏ **What's the pathway after this course?**

The two natural next steps are machine learning (see AI & ML Engineering — 12 Weeks) and SQL for data analysis. You'll leave with a specific recommendation based on your goals.

About Prasant Mishra



Prasant Mishra

Founder & Instructor — Qurious Academy

GenAI Solutions Architect

15+ Yrs Teaching

2500+ Students

60+ Projects Delivered

Prasant has used NumPy, Pandas, and the Python data science stack extensively across projects — from cleaning and analysing datasets for AI training pipelines to building data workflows for client projects. He knows where beginners get stuck (Pandas indexing, the groupby mental model, matplotlib's figure/axes architecture) and has designed the course to prevent those sticking points.

His teaching philosophy for data science: you learn data wrangling by wrangling messy data, not clean tutorial datasets. Every project in this course uses data that requires real decisions — which is why the skills transfer to your actual work.

QuriousAcademy

Focused, live learning for curious minds. No fluff — just the concepts that matter.

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Rolling start dates

₹5,999

Enroll at quriousacademy.com

or visit quriousacademy.com/courses

Questions? Reach us at hello@quriousacademy.com

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